

# HealthCore Medical Reference Guide

## 1. Hypertension (High Blood Pressure)

Hypertension is defined as a systolic blood pressure consistently above 140 mmHg or diastolic pressure above 90 mmHg. It is a major risk factor for heart disease, stroke, and kidney failure. Primary hypertension has no identifiable cause and accounts for 90-95% of cases. Secondary hypertension is caused by an underlying condition such as kidney disease or hormonal disorders.

### Treatment Options:

First-line medications include ACE inhibitors (e.g., Lisinopril), ARBs (e.g., Losartan), calcium channel blockers (e.g., Amlodipine), and thiazide diuretics. Lifestyle changes such as reducing sodium intake to less than 2g/day, regular aerobic exercise of 30 minutes 5 days a week, weight loss, and limiting alcohol are strongly recommended.

## 2. Type 2 Diabetes Mellitus

Type 2 diabetes is characterized by insulin resistance and relative insulin deficiency. Fasting blood glucose  $\geq 126$  mg/dL or HbA1c  $\geq 6.5\%$  confirms the diagnosis. It affects approximately 10% of the global adult population and is strongly associated with obesity, physical inactivity, and genetic predisposition.

### Management Protocol:

Metformin is the preferred initial pharmacological agent. Target HbA1c is below 7% for most patients. Second-line agents include SGLT2 inhibitors, GLP-1 receptor agonists, DPP-4 inhibitors, and sulfonylureas. Patients should monitor fasting blood glucose daily and undergo HbA1c testing every 3 months.

## 3. Common Drug Interactions

| Drug A      | Drug B                | Interaction             | Severity |
|-------------|-----------------------|-------------------------|----------|
| Warfarin    | Aspirin               | Increased bleeding risk | High     |
| Metformin   | Alcohol               | Lactic acidosis risk    | Moderate |
| Lisinopril  | Potassium supplements | Hyperkalemia            | High     |
| Simvastatin | Clarithromycin        | Myopathy risk           | High     |
| Amlodipine  | Grapefruit juice      | Increased drug level    | Moderate |

## 4. Emergency Protocols

Anaphylaxis: Administer Epinephrine 0.3mg IM immediately in the outer thigh. Follow with antihistamines and corticosteroids. Monitor for biphasic reaction for 4-6 hours. Cardiac Arrest: Begin

CPR at 100-120 compressions/minute. Use AED as soon as available. Administer Epinephrine 1mg IV every 3-5 minutes during resuscitation.